Emre Küçükgöçer

Computer Engineering Student | Software & AI Intern Candidate

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PROFESSIONAL SUMMARY

Fourth-year Computer Engineering student specializing in Computer Vision, LLMs, and real-time software architectures. Proven ability to design end-to-end systems managed with Git, from training YOLO-based Edge AI models to orchestrating multi-agent workflows with LangGraph and implementing RAG architectures. Skilled in building multi-tenant, zero-latency mobile interfaces using Flutter, Firebase, and FastAPI. Passionate about solving real-world engineering problems through hardware-software integration and eager to contribute to mission-critical, autonomous, and decision-support systems.

SKILLS

AI, LLMs & Computer Vision: Large Language Models, API Integration, Prompt Engineering, Multi-Agent Systems, LangGraph, RAG Architectures, YOLO Object Detection, Edge AI, TensorFlow Lite, Data Augmentation, Roboflow, Image Processing, OpenCV, Data Analysis, Pandas, NumPy

Software Architecture & Backend: FastAPI, Flutter, Dart, Firebase, NoSQL, Real-Time Data Synchronization, Multi-Tenant Architecture, RESTful APIs

Programming Languages & Tools: Python, Dart, Java, SQL, Git, GitHub

PROJECTS

DentAI - 1st Place, TAÜ Hackathon 2026

- Architected and deployed an end-to-end, privacy-preserving MedTech mobile platform using Flutter to accelerate clinical triage through real-time dental anomaly detection.
- Engineered an Edge AI architecture by deploying a TensorFlow Lite model directly on-device, enabling strict data localization and instant decision support without cloud dependency.
- Trained a custom YOLOv8-Nano object detection model and applied hard negative mining for braces and tartar cases to suppress false positives, achieving 88.8% accuracy in a 50-user field test.
- Established an optical preprocessing pipeline with OpenCV CLAHE for algorithmic noise reduction and a custom 3D-printed macro apparatus for hardware-level image stabilization.
- Integrated the backend into a state-driven Flutter UI with autonomous target-locking based on a confidence threshold above 80% and automated clinical background reporting.

Health Management System - TÜBİTAK Funded Project

- Developed a secure, multi-tenant cross-platform application using Flutter to support isolated and real-time communication between doctors and patients.
- Engineered a NoSQL database structure with Firebase Realtime Database and implemented Rule-Based Access Control to ensure data isolation across multiple clinics.
- Resolved critical memory bottlenecks by refactoring heavy media payloads from Base64 to Firebase Cloud Storage URLs, significantly reducing socket load.
- Designed an asynchronous data queueing mechanism for network disruptions and implemented secure local storage authentication with custom clinic invite codes.
- Executed a background data migration script to adapt legacy users to a new many-to-many architecture without data loss.

Council AI - Multi-Agent Educational Assistant

- Architected an end-to-end educational ecosystem using Clean Architecture principles, separating data, domain, and presentation layers within a Flutter 3 mobile environment.
- Engineered a sequential LangGraph pipeline using Google Gemini 2.0 Flash and designed three specialized AI agents: Socrates, Fact Checker, and Quest Master.
- Designed the agent system to academically analyze, strictly verify to prevent hallucinations, and pedagogically gamify complex queries.
- Developed an asynchronous RESTful API with FastAPI and Uvicorn to manage the AI execution environment and ensure low-latency client-server communication.
- Implemented Riverpod StateNotifier for scalable state management and designed a custom dark-neon glassmorphism interface with flutter_animate for real-time user experience transitions.

EXPERIENCE

Ratione Divide (Radia AI) - Medical AI R&D Intern **Sep 2026 - Present**

- Contributing to medical AI product engineering, with a focus on explainable clinical decision support and medical-imaging workflows.
- Supporting the path from model behavior to usable health-product features through data preparation, model evaluation, and application integration.

HT Bilişim - Applied AI & ML Engineering Intern **Aug 2026 - Sep 2026**

- Spearheaded the design of a production-grade Enterprise RAG architecture, leveraging advanced semantic chunking, embedding models, and Vector Databases (Pinecone/Milvus) to ensure highly accurate, hallucination-free context retrieval for proprietary datasets.
- Orchestrated autonomous Multi-Agent AI workflows via LangGraph and LangChain, enabling dynamic tool-calling, multi-step reasoning, and strict output evaluation frameworks for complex decision-support systems.
- Established an end-to-end MLOps pipeline by containerizing large language models and inference endpoints with Docker, bridging the gap between local R&D and scalable, drift-monitored production deployments.

Uçak Yazılım - Software Engineering Intern **Jul 2026 - Aug 2026**

- Engineered a high-throughput, distributed OSINT (Open Source Intelligence) data pipeline utilizing Python and asynchronous workers to scrape, sanitize, and aggregate high-volume, multi-source unstructured web data in real-time.
- Architected a zero-latency microservice layer via FastAPI to serve processed intelligence data, significantly optimizing data retrieval speeds and ensuring seamless RESTful integration for analytical front-end consumption.
- Designed complex ETL (Extract, Transform, Load) workflows to normalize and index raw intelligence into a centralized, query-optimized relational database, completely eliminating data redundancy and accelerating complex search queries.

Yandex - SEO **Feb 2025 - Mar 2025**

- Conducted keyword research, performance tracking, and technical SEO audits to improve organic visibility and site structure.

EDUCATION & CERTIFICATIONS

Sivas Cumhuriyet University | Computer Engineering **Sep 2023 - Jun 2027**

Samsung Innovation Campus | Artificial Intelligence & Marketing **July 2026 - Sept 2026**